

Five-year outcomes following timely primary percutaneous intervention, late primary percutaneous intervention, or a pharmaco-invasive strategy in ST-segment elevation myocardial infarction: the FAST-MI programme

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Aims

ST-segment elevation myocardial infarction (STEMI) guidelines recommend primary percutaneous coronary intervention (pPCI) as the default reperfusion strategy when feasible ≤ 120 min of diagnostic ECG, and a pharmaco-invasive strategy otherwise. There is, however, a lack of direct evidence to support the guidelines, and in real-world situations, pPCI is often performed beyond recommended timelines. To assess 5-year outcomes according to timing of pPCI (timely vs. late) compared with a pharmaco-invasive strategy (fibrinolysis with referral to PCI centre).

Methods and results

The French registry of Acute ST-elevation and non-ST-elevation Myocardial Infarction (FAST-MI) programme consists of nationwide observational surveys consecutively recruiting patients admitted for acute myocardial infarction every 5 years. Among the 4250 STEMI patients in the 2005 and 2010 cohorts, those with reperfusion therapy and onset-to-first call time < 12 h ($n = 2942$) were included. Outcomes at 5 years were compared according to type of reperfusion strategy and timing of pPCI, using Cox multivariable analyses and propensity score matching. Among those, 1288 (54%) patients had timely pPCI (≤ 120 min from ECG), 830 (28%) late pPCI (> 120 min), and 824 (28%) intravenous fibrinolysis. Five-year survival was higher with a pharmaco-invasive strategy (89.8%) compared with late

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pPCI [79.5%; adjusted hazard ratio (HR) 1.51; 1.13–2.02] and similar to timely pPCI (88.2%, adjusted HR 1.02; 0.75–1.38). Concordant results were observed in propensity score-matched cohorts and for event-free survival.

Conclusion

A substantial proportion of patients have pPCI beyond recommended timelines. As foreseen by the guidelines, these patients have poorer 5-year outcomes, compared with a pharmaco-invasive strategy.

Keywords

Primary PCI • Fibrinolysis • Acute myocardial infarction • Timing • Long-term outcome

Introduction

Primary percutaneous coronary intervention (pPCI) is the default strategy for treating ST-segment elevation myocardial infarction (STEMI) at the acute stage. The guidelines, however, recommend a pharmaco-invasive strategy (intravenous fibrinolysis followed by transfer to a PCI centre for secondary coronary angiography $\pm$ PCI) when time from diagnostic ECG to pPCI exceeds 120 min.¹ In everyday life, however, a substantial proportion of pPCI procedures are performed beyond recommended timelines, and no randomized comparison exists between the pharmaco-invasive strategy and late pPCI. The purpose of the present study was to assess 5-year outcomes in patients receiving reperfusion therapy for ST-segment elevation acute myocardial infarction (AMI), comparing patients receiving timely pPCI, late (>120 min from diagnostic ECG) pPCI, or a pharmaco-invasive strategy, using data from the French registry of Acute ST-elevation and non-ST-elevation Myocardial Infarction (FAST-MI) 2005 and 2010 cohorts.^{2,3}

Methods

Population

Consecutive patients included in the FAST-MI 2005 and 2010 registries, presenting with ST-segment elevation within 12 h of symptom onset and receiving reperfusion therapy with either pPCI or fibrinolytic treatment, were included.

FAST-MI 2005 and 2010 registries

The population and methods of the FAST-MI registries have been previously described in detail.^{2–4} Briefly, the objective of these observational prospective studies was to collect comprehensive data on the management and outcomes of consecutive patients admitted to intensive care units for definite AMI over a 1-month period in France, irrespective of the type of institution receiving the patients (i.e. university hospitals, public hospitals or private clinics, with or without on-site catheterization facilities). Patients ≥ 18 years of age admitted ≤ 48 h of symptom onset were included if they had: (i) elevated serum markers of myocardial necrosis higher than twice the upper limit of normal and (ii) either symptoms compatible with AMI and/or electrocardiographic changes on at least two contiguous leads with pathologic Q waves (at least 0.04 s) and/or persisting ST-elevation or depression >0.1 mV. Patients were excluded if they had iatrogenic AMI (in particular, Types 4 and 5 AMI), or if the diagnosis of MI was invalidated in favour of an alternative.

Baseline characteristics (demographics, risk factors, and medical history) were collected prospectively. All data were recorded on computerized case record forms by dedicated research technicians sent in each of the centres at least once a week.

In 2005 (clinicaltrials.gov identifier: NCT00673036), 3670 patients were recruited in 223 centres (60% of the centres treating patients with AMI), and in 2010 (clinicaltrials.gov identifier: NCT01237418), 4169 patients were recruited in 213 centres (76% of the centres treating AMI patients). Patients were recruited over a 1-month period; in 2005 diabetic patients were recruited for one additional month; in 2010, voluntary centres could extend recruitment for up to one additional month.

The protocols of FAST-MI 2005 and FAST 2010 were reviewed by the Committee for the Protection of Human Subjects in Biomedical Research of Saint Antoine University Hospital and Saint-Louis University Hospital, respectively. All patients gave informed consent for their participation in the study.

Follow-up data

For both cohorts, centralized follow-up was organized at the French Society of Cardiology and performed by dedicated research technicians of Unité de recherche Clinique de l'Est Parisien (URC-EST). Hospital discharge reports were sought for each reported event leading to hospitalization or death and analysed by at least one physician from the research team; all cases of cardiovascular events in which the final diagnosis appeared debatable were reviewed by a three-member critical events committee (N.D., P.G., and T.S.). Vital status at 5 years could be assessed in 97% of FAST-MI 2005 and $>95\%$ of FAST-MI 2010 patients. Morbidity (recurrent MI or ischaemic stroke) at 5 years could be collected in 86% of the FAST-MI 2005 and 79% of the FAST-MI 2010 patients; event-free survival was analysed on the populations with morbidity follow-up available.

Current population

For the present study, patients with ST-segment elevation AMI, presenting within 12 h of symptom onset and receiving either fibrinolytic therapy or pPCI were included. Three populations were compared: patients with pPCI performed within 120 min of diagnostic ECG ('timely pPCI', $n = 1288$), patients with pPCI beyond 120 min of diagnostic ECG ('late pPCI', $n = 830$), and pharmaco-invasive strategy ($n = 824$ with fibrinolytic therapy, 98% undergoing coronary angiography, 86% actually undergoing pPCI).

Statistical analysis

Continuous variables are reported as means and standard deviations or medians and interquartile ranges, when appropriate. Discrete

variables are described as counts and percentages. Comparisons were made with the χ^2 or Fisher's exact tests for discrete variables, and by unpaired *t* tests, Mann–Whitney tests, or analysis of variance for continuous variables.

Crude and event-free survival curves were estimated using the Kaplan–Meier estimation and compared using log-rank tests. To assess the independent association of type of reperfusion therapy with outcomes, three analyses were performed: (i) a multivariable backward stepwise Cox analysis using the Global Registry of Acute Coronary Events (GRACE) score as covariable; (ii) a full multivariable analysis with the following covariables: baseline characteristics (age, gender, risk factors, past cardiovascular history, and associated conditions), medications used before the index AMI, study year, time from symptom onset to first call, AMI location, and out-of-hospital cardiac arrest; (iii) separate multivariable Cox analyses using propensity scores for late pPCI vs. pharmaco-invasive strategy, and for timely pPCI vs. pharmaco-invasive strategy (see below). For event-free survival analyses, only those patients with full morbidity follow-up available were included in the analyses. Forward models were also used to check the results' robustness. In addition, propensity score matching was used to compare the pharmaco-invasive strategy with either late pPCI or timely pPCI.

Binary logistic regression analysis was used to create propensity scores to balance baseline characteristics in patients treated with fibrinolysis and either late pPCI or timely pPCI. Two propensity scores were calculated, one for comparing fibrinolysis with timely pPCI and the other to compare fibrinolysis with late pPCI. Cohorts of patients were built, matched on the propensity scores using a greedy matching procedure (1 to 1 matching, two groups of 549 patients comparing late pPCI to fibrinolysis, and two groups of 562 patients comparing fibrinolysis with timely pPCI). Variables used for calculating the propensity score included all baseline characteristics, as well as study year, infarct location, and time from onset to first call. For the comparison with late pPCI, the standardized difference was -80% before matching and -0.50% after matching; for the comparison of the pharmaco-invasive strategy with timely pPCI, the standardized difference was -105% before matching and -0.08% after matching.

Statistical analyses were performed using IBM SPSS 25.0 (IBM SPSS Inc.) and NCSS 12 (NCSS, LLC, Kaysville, UT, USA). For all analyses, a two-sided *P*-value <0.05 was considered to be statistically significant.

Results

Study population

From a total of 4250 patients with STEMI, 3068 had reperfusion therapy with primary PCI or fibrinolysis, among whom 2942 had called within 12 h of symptom onset. Among those, 1288 (44%) underwent timely pPCI, 830 (28%) late pPCI, and 824 (28%) had a pharmaco-invasive strategy (TNK-TPA 84%). Patients in the late pPCI group were older, with more risk factors (except current smoking) and comorbidity, with a higher GRACE risk score and more frequently anterior MI than timely pPCI or fibrinolysis (Table 1).

Initial management and medications in the first 48 h

Patients with timely pPCI or a pharmaco-invasive strategy more frequently called the emergency medical system, and 63% of fibrinolytic-

treated patients received pre-hospital lysis. Aspirin, beta-blockers, statins, and low molecular weight heparin were used in similar proportions in the three groups; clopidogrel was used more often in fibrinolytic-treated patients. More patients in the late pPCI group received diuretics after hospital admission, and fewer patients in the pharmaco-invasive group received prasugrel (Table 2).

In-hospital complications and early mortality

In-hospital complications are described in Table 2. There was no significant difference in bleeding complications, although intracranial bleedings were numerically more frequent in the pharmaco-invasive group; in patients aged ≥ 75 years, the intracranial bleeding rate was 0.8% for the pharmaco-invasive strategy (5% received half-dose lytics).

Five-year outcomes

The propensity score-matched cohorts were well balanced (Supplementary material online, Tables S1 and S2). Early management and in-hospital events are presented in Supplementary material online, Tables S3 and S4. In the matched cohorts, 5-year survival was 89.3% (86.8–91.8%) for the pharmaco-invasive strategy and 81.8% (78.5–85.1%) for late pPCI [hazard ratio (HR) late pPCI vs. pharmaco-invasive: 1.77, 95% confidence interval (CI): 1.28–2.44; *P* = 0.001], and 89.3% (86.8–91.8%) for both the pharmaco-invasive and the timely pPCI groups (HR 1.01, 95% CI: 0.70–1.44; *P* = 0.97) (Figure 1). The respective figures for the combined endpoint of death, AMI or stroke at 5 years were 85.2% (82.0–88.3%) and 76.5% (72.8–80.2%) (HR late pPCI vs. pharmaco-invasive: 1.64, 1.23–2.18; *P* = 0.001) and 85.1% (82.0–88.2%) vs. 84.3% (81.2–87.6%) (HR timely pPCI vs. pharmaco-invasive: 1.08, 0.79–1.47; *P* = 0.64).

In the whole population, 5-year crude survival was 89.8% (87.6–92.0%) for the pharmaco-invasive strategy, 88.2% (86.4–90.0%) for timely pPCI, and 79.5% (76.8–82.2%) for late pPCI (Figure 2). Using the pharmaco-invasive strategy as the reference group, the HR and 95% CI were 1.14 (0.86–1.49) for timely pPCI and 1.78 (1.36–2.33) for late pPCI after adjustment on the GRACE score, and 1.04 (0.78–1.37) and 1.47 (1.11–1.95), respectively after full adjustment. After adjustment on the propensity scores, the HR for timely pPCI was 1.02 (0.75–1.38) and 1.51 (1.13–2.02) for late pPCI, compared with the pharmaco-invasive strategy. For the combined endpoint of death, AMI or stroke at 5 years, the fully adjusted HR for timely pPCI was 1.11 (0.87–1.42, *P* = 0.40) and 1.46 (1.14–1.88, *P* = 0.003) for late pPCI, compared with the pharmaco-invasive strategy (Take home figure).

Subgroup analyses

There was no interaction among subgroups according to survey year, age, sex, location of AMI, initial call to emergency medical services, and time from onset to first call, when comparing 5-year mortality in the pharmaco-invasive and late pPCI groups, after adjustment on the propensity score (Figure 3). Of note, 5-year mortality did not differ between the two reperfusion strategies in patients calling late (>90 min) after symptom onset (HR 1.01, 95% CI: 0.60–1.69; *P* for interaction 0.07).

Finally, 5-year mortality did not differ between late fibrinolysis (i.e. >30 min from diagnostic ECG, as defined by the STEMI guidelines in

Table 1 Baseline characteristics according to type of reperfusion strategy and timing of primary percutaneous coronary intervention

	Timely primary PCI (n = 1288)	Late primary PCI (n = 830)	Pharmaco-invasive (n = 824)	P-value
Demography, risk factors, and medical history				
Age (years), mean $\pm$ SD	61 $\pm$ 13	64 $\pm$ 15	60 $\pm$ 13	<0.001
Age $\geq$ 75 years	244 (18.9)	212 (25.5)	125 (15.2)	<0.001
Women	282 (21.9)	250 (30.1)	173 (21.0)	<0.001
BMI $\geq$ 30 Kg/m ²	245 (20.1) (n = 1220)	170 (21.9) (n = 778)	163 (21.2) (n = 770)	0.621
Diabetes	213 (16.5)	195 (23.5)	142 (17.2)	<0.001
Hypertension	552 (42.9)	423 (51.0)	347 (42.7)	<0.001
Current smoking	582 (45.2)	324 (39.0)	391 (47.5)	0.001
Hypercholesterolaemia	524 (40.7)	348 (41.9)	337 (40.9)	0.843
Family history of CAD	346 (26.9)	210 (25.3)	259 (31.4)	0.014
Prior AMI	136 (10.6)	89 (10.7)	64 (7.8)	0.065
Prior PCI	139 (10.8)	80 (9.6)	53 (6.4)	0.003
Prior stroke/TIA	53 (4.1)	40 (4.8)	12 (1.5)	<0.001
Peripheral artery disease	47 (3.7)	51 (6.1)	32 (3.9)	0.017
History of heart failure	16 (1.2)	25 (3.0)	3 (0.4)	<0.001
Chronic kidney disease	30 (2.3)	16 (1.9)	12 (1.5)	0.368
Chronic obstructive lung disease	71 (5.5)	34 (4.1)	22 (2.7)	0.007
History of cancer	88 (6.8)	67 (8.1)	39 (4.7)	0.021
Initial management				
Time from onset to first call (min), median (IQR)	60 (30–150)	70 (30–210)	50 (20–105)	<0.001
EMS as first contact	784 (60.9)	329 (39.6)	497 (60.3)	<0.001
Transfer to catheterization laboratory	99 (7.7)	272 (32.2)	267 (32.4)	<0.001
Initial presentation				
Anterior infarction	527 (40.9)	350 (42.2)	297 (36.0)	0.024
GRACE risk score	141 $\pm$ 32 (n = 1241)	147 $\pm$ 35 (n = 778)	141 $\pm$ 31 (n = 770)	<0.001
Killip class $\geq$ 2	135 (10.6)	140 (17.1)	94 (11.5)	<0.001
Out-of-hospital cardiac arrest	29 (2.3)	16 (1.9)	23 (2.8)	0.496

AMI, acute myocardial infarction; BMI, body mass index; CAD, coronary artery disease; EMS, emergency medical services; IQR, interquartile range; SD, standard deviation; TIA, transient ischaemic attack.

use at the time) and late pPCI (adjusted HR 1.24, 95% CI: 0.82–1.87; $P = 0.298$); in fibrinolytic-treated patients, there was no difference between those treated in the ambulance or in spoke hospitals (adjusted HR 1.18, 95% CI: 0.68–2.04). Only 2% of the patients received half-dose fibrinolytic treatment (4% in those aged 75 years or more); the dominant pharmaco-invasive strategy, received by 37% of the patients, combined full-dose TNK, clopidogrel loading dose (≥ 300 mg) and low molecular weight heparin; 5-year mortality did not differ between those receiving the dominant combination and other types of pharmaco-invasive strategy combinations (9.5% vs. 10.6%); TIMI major bleeding was 2.3% vs. 1.7%, and rates of intracranial bleeding were 0.7% vs. 0.4%, respectively.

Discussion

In these French nationwide surveys of patients hospitalized for STEMI, 28% of the population had a pPCI beyond the recommended timelines (>120 min from diagnostic ECG). Patients receiving a

pharmaco-invasive strategy at the acute stage had improved 5-year survival and event-free survival, compared with pPCI performed beyond the recommended timelines. In contrast, 5-year death and death or major cardiovascular events did not differ between timely pPCI and the pharmaco-invasive strategy groups. To the best of our knowledge, these results are the first directly comparing 5-year outcomes according to different modalities of reperfusion strategy and analysed by timing of pPCI.

Following the results of the initial trials comparing stand-alone fibrinolytic therapy with primary PCI⁵ and the ensuing guidelines,³ the use of primary PCI has consistently increased in the past two decades, with a concomitant decline in the use of fibrinolytic therapy in many countries, including France where it decreased from 40% in 2005 to 19% in 2010.^{6–8} However, cardiologists and emergency physicians may have tended to forget that achieving timely performance of pPCI is often more difficult than expected, with proportions of patients treated more than 120 min after the diagnostic ECG ranging from 25% to as much as 50%, according to country and organization of care.^{6,7,9–11} Many factors, including patient age, time of onset,

Table 2 Early management and in-hospital events

	Timely primary PCI (n = 1288)	Late primary PCI (n = 830)	Pharmaco-invasive (n = 824)	P-value
Early management				
Pre-hospital vs. in-hospital lysis	—	—	522 (63.3)/302 (36.7)	—
Time from onset to needle (min), median (IQR)	—	—	120 (80–189)	—
Aspirin	1263 (98.1)	813 (98.0)	806 (97.8)	0.928
Clopidogrel	1056 (82.0)	748 (90.1)	761 (92.4)	<0.001
Prasugrel	422 (32.8)	169 (20.4)	107 (13.0)	<0.001
Glycoprotein IIb/IIIa inhibitors	477 (53.8)	232 (46.8)	38 (11.5)	<0.001
Unfractionated heparin	836 (64.9)	492 (59.3)	537 (65.2)	0.015
LMWH	838 (65.1)	534 (64.3)	556 (67.5)	0.363
Fondaparinux	121 (9.4)	86 (10.4)	37 (4.5)	<0.001
Bivalirudin	64 (5.0)	28 (3.4)	3 (0.4)	<0.001
ACE-inhibitor or ARB	843 (65.5)	513 (61.8)	463 (56.2)	<0.001
Diuretic	264 (20.5)	230 (27.7)	144 (17.5)	<0.001
Beta-blocker	1056 (82.0)	664 (80.0)	671 (81.4)	0.514
Statin	1156 (89.8)	739 (89.0)	720 (87.4)	0.236
Pre-PCI TIMI Flow 2 or 3	369 (30.4) (n = 1214)	246 (31.6) (n = 779)	455 (59.8) (n = 761)	<0.001
Post-PCI TIMI Flow 2 or 3	1126 (95.3) (n = 1182)	661 (92.2) (n = 717)	613 (89.6) (n = 684)	<0.001
Rescue PCI	—	—	256 (34.4) (n = 745)	—
In-hospital complications				
Recurrent AMI	10 (0.8)	9 (1.1)	19 (2.3)	0.08
Stroke	5 (0.4)	5 (0.6)	9 (1.1)	0.141
Intracranial bleeding	1 (0.1)	1 (0.1)	4 (0.5)	0.105
TIMI major bleeding	17 (1.3)	23 (2.8)	16 (1.9)	0.058
TIMI minor bleeding	34 (2.6)	17 (2.0)	20 (2.4)	0.687
Blood transfusion	41 (3.2)	32 (3.9)	27 (3.3)	0.688
Third degree atrioventricular block	19 (1.5)	27 (3.3)	12 (1.5)	0.007
Atrial fibrillation	97 (7.5)	97 (11.7)	73 (8.9)	0.005
Secondary cardiogenic shock	24 (1.9)	28 (3.4)	19 (2.3)	0.084
LVEF ≤40%	207 (17.7) (n = 1170)	161 (21.3) (n = 757)	118 (15.9) (n = 742)	0.022
In-hospital death	27 (2.1)	36 (4.3)	25 (3.0)	0.013
30-day death	32 (2.5)	36 (4.3)	27 (3.3)	0.062

ACE, angiotensin-converting enzyme; AMI, acute myocardial infarction; ARB, angiotensin receptor blocker; IQR, interquartile range; LMWH, low molecular weight heparin; LVEF, left ventricular ejection fraction; TIMI, thrombolysis in myocardial infarction.

or geographical location may impact timing of pPCI.¹² In such patients, a pharmaco-invasive strategy may be an attractive therapeutic option and is actually recommended in the absence of specific contra-indications, with initial (preferably pre-hospital) fibrinolysis followed by transportation to a PCI centre capable of providing rescue¹³ or routine PCI in the absence of fibrinolysis failure or recurrent symptoms.^{3,14–17} To date, however, there is no strong evidence supporting this recommendation, which is essentially based on the decreasing clinical efficacy of pPCI with increasing times from symptom onset.¹⁸

In a real-world setting, the FAST-MI 2005 registry previously showed that long-term mortality was not different in patients with a pharmaco-invasive strategy and in pPCI-treated patients,^{4,19} and was actually associated with improved survival when prehospital fibrinolysis was used. The STRategic Reperfusion Early After Myocardial Infarction (STREAM) study documented a 14% lower risk of the 30-

day primary endpoint with a pharmaco-invasive strategy and primary PCI in patients presenting within 3 h of symptom onset and in whom pPCI could not be performed in <60 min, a difference not statistically significant.²⁰ Similar results were found at 1 year.²¹ In the trial, however, ~25% of the patients actually had sheath insertion in <1 h from randomization.

Our current findings strongly support current guidelines. At the acute stage, compared with a late pPCI strategy, the pharmaco-invasive strategy was associated with a reduced risk of atrioventricular block, and numerically lower risk of in-hospital or 30-day death. As in the Korean registry,¹⁰ there was no excess risk of bleeding, a finding possibly related to the short half-life of tenecteplase, which was the most commonly used fibrinolytic treatment in our population,²² and to the preferential use of the radial access (81%). In the propensity score-matched cohorts, two intracranial bleedings were observed in the pharmaco-invasive cohort vs. one in the late pPCI

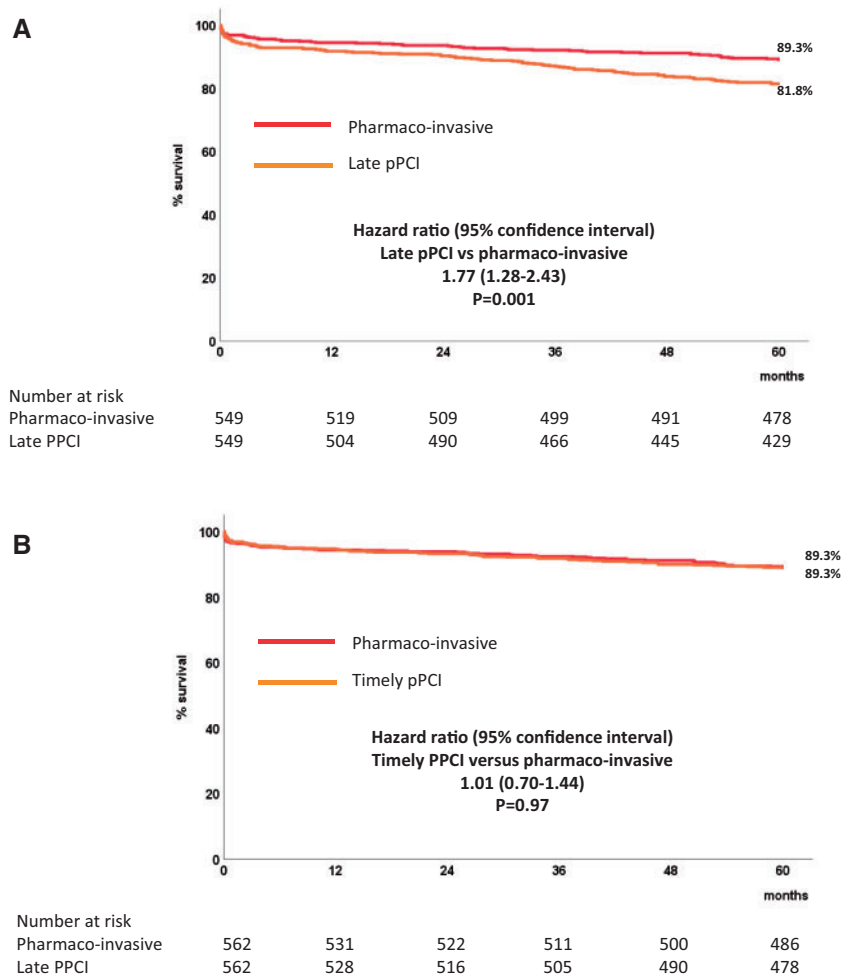


Figure 1 Five-year survival in propensity score-matched populations. (A) pharmaco-invasive strategy versus late primary percutaneous coronary intervention (pPCI); (B) pharmaco-invasive strategy versus timely pPCI. PPCI, primary percutaneous coronary intervention.

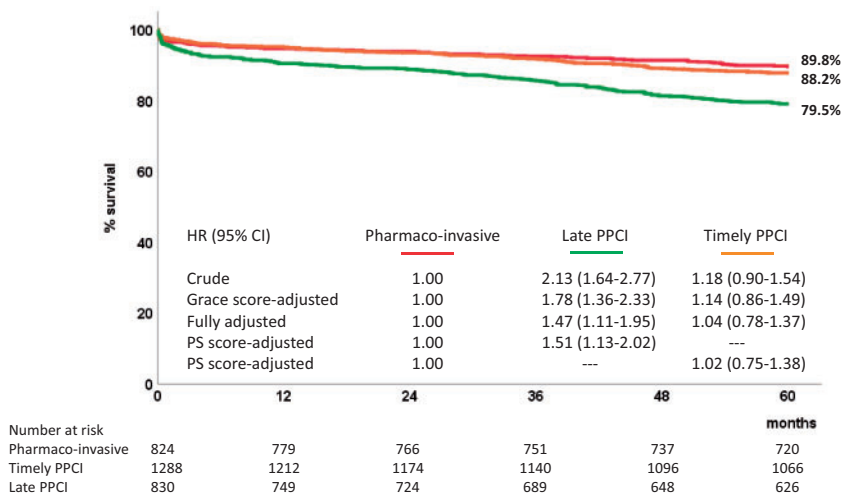
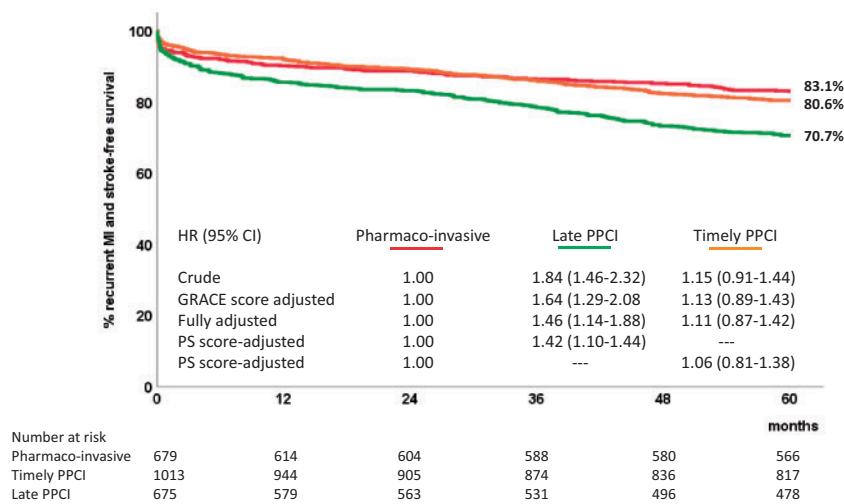


Figure 2 Five-year survival according to type of reperfusion strategy and timing of primary percutaneous coronary intervention. CI, confidence interval; HR, hazard ratio; pPCI, primary percutaneous coronary intervention.



Take home figure Five-year acute myocardial infarction-free and stroke-free survival according to type of reperfusion strategy and timing of primary percutaneous coronary intervention. CI, confidence interval; HR, hazard ratio; PPCI, primary percutaneous coronary intervention.

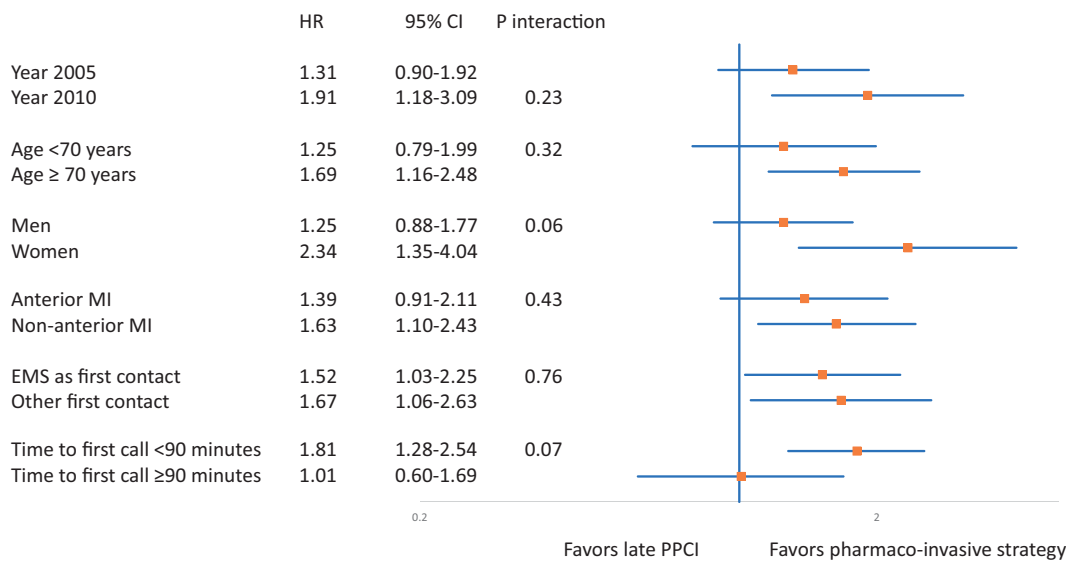


Figure 3 Adjusted hazard ratios and 95% confidence intervals for 5-year mortality among subgroups: pharmaco-invasive strategy versus late primary percutaneous coronary intervention (PPCI). EMS, emergency medical system; MI, myocardial infarction.

cohort. Overall, the risk of stroke was similar in the two cohorts. There was also a numerically increased risk of recurrent AMI (2.7% vs. 1.3%, $P = 0.13$), a figure concordant with the rate of recurrent AMI in the STREAM trial (2.5% in the pharmaco-invasive arm and 2.2% in the pPCI arm).²⁰ At 5 years, however, both mortality and the combined endpoint of death, recurrent MI or stroke was lower in patients receiving a pharmaco-invasive strategy, compared to late pPCI.

Timing of reperfusion therapy is crucial, particularly in patients treated with intravenous fibrinolysis, as older thrombi are more difficult to lyse.^{18,23,24} Recently, the Early Routine Catheterization After

Alteplase Fibrinolysis vs. Primary PCI in Acute ST-Segment-Elevation Myocardial Infarction (EARLY-MYO) showed that epicardial and myocardial reperfusion was improved in patients seen early and pre-treated with half-dose alteplase, compared to delayed pPCI.²⁵ In this regard, although the interaction was not significant ($P = 0.07$), the subgroup analysis according to time from onset to first call suggests that in patients seen late after the onset of AMI, the advantage of a pharmaco-invasive strategy over late pPCI may be lost. In such patients, late pPCI may remain the preferable option, as it minimizes the risk of intracranial bleeding.

This study is subject to the limitations inherent in all observational studies; namely, no direct causality can be formally demonstrated. Use of multivariable analyses and propensity score matching, however, takes into consideration bias related to the potential confounders that were collected in the survey. In this regard, the snapshot nature of the registries allows to collect a considerable number of items, and therefore to increase the quality of adjustments. Finally, our data come from a country with an extremely developed prehospital emergency care system (Service d'Aide Médicale Urgente, SAMU), with emergency physicians on board the ambulance, deciding direct transport to PCI capable hospitals for pPCI, or delivery of a pharmaco-invasive strategy; our data, therefore, may not be generalized to different systems of care.

Conclusion

Too many STEMI patients still undergo primary PCI later than recommended. Efforts should be made to ensure that primary PCI is performed within 120 min of diagnostic ECG, and as early as possible after symptom onset. If these timelines cannot be achieved, a pharmaco-invasive strategy should be instituted, as it may increase long-term survival and event-free survival, compared to a late primary PCI strategy, therefore, lending support to current guidelines. The current results should be an incentive for cardiologists and emergency physician to realistically assess the expected times to performance of primary PCI, in order to choose alternative strategies when the recommended timelines are unlikely to be met.

Supplementary material

Supplementary material is available at *European Heart Journal* online.

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